Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mass” and “system” to explain your model or result. Sentence starters are optional.

mass: a measure of how strongly an object resists acceleration

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A car starts at +24 m/s and brakes at -4 m/s² until rest. A student extends that acceleration to 8 s.
Find the stop time/distance and explain why extension is invalid for ordinary braking.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

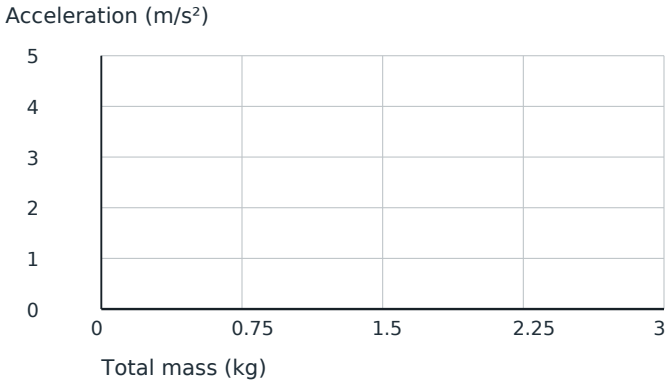
02 Worked example: explain the model

Reference: $a = F_{net}/M = m_h\, g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials.
Label what was held constant.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Test an inverse relationship

The table contains ideal data for a constant 0.49 N net force.
Calculate 1/M for each mass.
Sketch acceleration versus 1/M, then state the slope's units and physical meaning.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

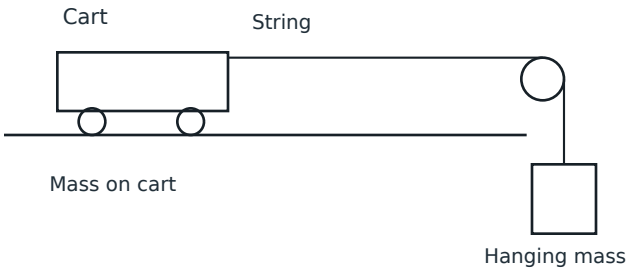
M (kg)	a (m/s²)
0.50	0.980
0.75	0.653
1.00	0.490
1.50	0.327

Marzano check - circle: 1 2 3

04 System boundary

The cart and hanger accelerate together.
Use the setup model to explain why both belong in the total moving mass and why string tension is internal to the combined system.
State two idealizations.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

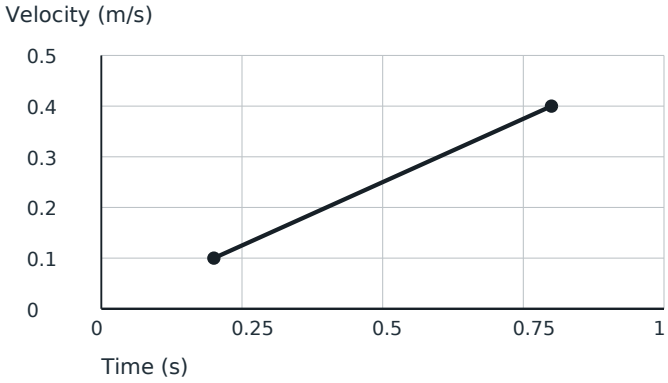


Marzano check - circle: 1 2 3

05 Measured slope and repetitions

A sensor records velocity changing from 0.10 m/s at 0.20 s to 0.40 m/s at 0.80 s.
Find the acceleration from the graph. Then design a table for three total masses and three trials per mass; label any data you actually measure.

Sentence starter: My design shows ... . It meets the requirement by ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 3 kg object hangs still from a vertical string.
Use g=9.8 N/kg.
Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram?
Draw the forces on the chosen object and label their sources.

Sentence starter: The … in my model represents … . Its label shows … .



Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

A 0.80 kg moving system accelerates at 0.60 m/s². At the same net force, predict a for 1.20 kg.
Explain how added resistance would change the prediction.

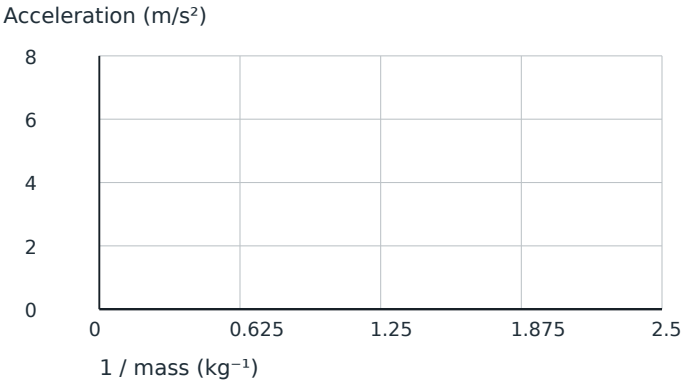
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A cart model predicts a=F/m. At constant net force 3 N, find accelerations for masses 0.5, 1, 2 kg.
Choose a transformed horizontal axis that makes a straight-line graph and state its slope.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: ① ② ③